

## Problem

A transmitter delivers maximum power to an antenna when the antenna is adjusted to represent a load of  $75\text{-}\Omega$  resistance in series with an inductance of  $4\text{ }\mu\text{H}$ . If the transmitter operates at  $4.12\text{ MHz}$ , find its internal impedance.

## Step-by-step solution

## Step 1 of 1

For maximum power transfer,

$$Z_L = Z_{th}^* \Rightarrow Z_i = Z_{th} = Z_L^*$$

$$Z_L = R + j\omega L = 75 + j(2\pi)(4.12 \times 10^6)(4 \times 10^{-6})$$

$$= 75 + j103.55\text{ }\Omega$$

$$Z_i = 75 - j103.55\text{ }\Omega .$$